

Sentiment Analysis of Movie Reviews by Genre

Computer-based Text Analysis / Summer Term 2025

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1. Introduction

The primary aim of this project is to explore how audiences perceive different movie genres by examining their written reviews through sentiment analysis. In the modern media environment, viewers frequently consult reviews before deciding to watch a film, while studios, critics, and streaming services seek insights into how genre affects audience satisfaction. Since reviews are usually unstructured text, analysing them effectively requires the use of Natural Language Processing (NLP) techniques.

The project focuses on the following research questions:

1. RQ1 – Positive Sentiment Percentage by Genre:

- Do reviews for certain movie genres (e.g., Comedy) tend to be more positive than others (e.g., Horror)?

2. RQ2 – Genre Consistency vs. Division in Audience Sentiment:

- Which genres show the most consistent audience sentiment, and which ones divide opinions?

3. RQ3 – Usefulness of Sentiment Analysis:

- Can sentiment analysis provide useful insights into audience preferences?

2. Literature Review & Background

Sentiment analysis is a common NLP technique for extracting subjective information from text, classifying opinions as positive, negative, or neutral. Previous research has shown that analyzing user reviews can provide insights into consumer satisfaction, content reception, and emotional engagement. Studies also suggest that genre affects audience reactions: light-hearted genres like Comedy often receive more positive feedback, while genres like Horror or Drama elicit more mixed or polarized responses.

Tools commonly used in sentiment analysis include:

- **TextBlob:** Provides polarity scoring between -1 (negative) and +1 (positive).
- **VADER (Valence Aware Dictionary for Sentiment Reasoning):** Optimized for social media text, effective for short and informal reviews.

3.Dataset & Data Collection

The dataset utilized for this project is the Movie Reviews and Emotion Dataset obtained from Kaggle, curated for research in sentiment analysis and text mining. It comprises a substantial collection of user-generated reviews across multiple movie genres, providing both textual and numerical data conducive to computational analysis. The dataset's multi-faceted structure supports exploration of audience sentiment patterns in relation to genre-specific content.

- **Source:** Kaggle ([Movie Reviews and Emotion Dataset](#))
- **Features:**

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 46173 entries, 0 to 46172
Data columns (total 8 columns):
#   Column          Non-Null Count  Dtype
---  -
0   Unnamed: 0      46173 non-null  int64
1   Ratings         46173 non-null  float64
2   Reviews         46173 non-null  object
3   movie_name      46173 non-null  object
4   Resenhas        46173 non-null  object
5   genres          46173 non-null  object
6   Description     46173 non-null  object
7   emotion         46173 non-null  object
dtypes: float64(1), int64(1), object(6)
memory usage: 2.8+ MB
```

- **Reviews:** The primary data consists of user-written textual reviews. These unstructured texts contain rich sentiment information, encompassing opinions, emotions, and evaluative judgments.
- **Ratings:** Each review is accompanied by a numeric rating ranging from 1 to 5, offering a quantitative measure of user satisfaction. This feature enables cross-validation of sentiment scoring techniques against explicit user ratings.
- **Genres:** Movies are categorized into one or more genres (e.g., Comedy, Horror, Drama), allowing for comparative analysis of sentiment trends across different thematic classifications. Multi-genre labelling is handled via list structures, which are later normalized for aggregation at the genre level.

Data Preprocessing

1. Cleaning Text Data:

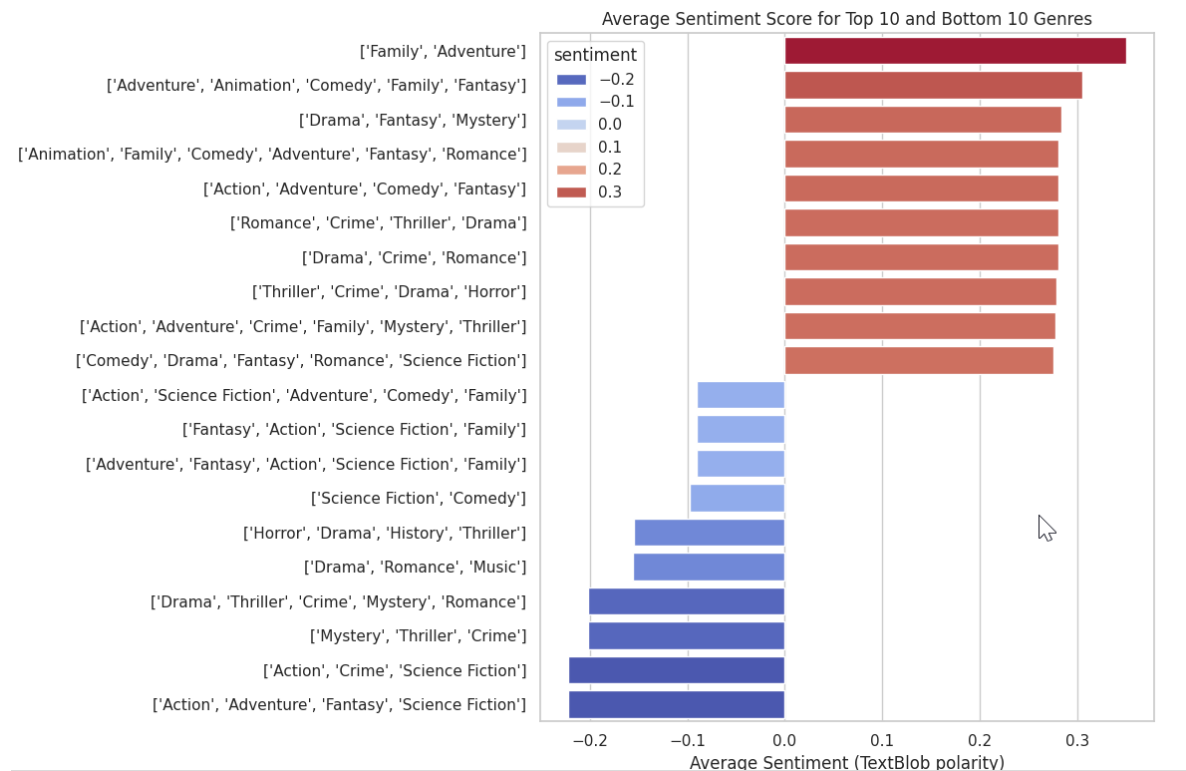
- Non-alphanumeric characters, punctuation, and stopwords were removed to reduce noise and improve the accuracy of sentiment scoring. Text normalization techniques, such as lowercasing and tokenization, were applied to standardize the data.

2. Multi-Genre Handling:

- Movies labelled with multiple genres were “exploded” so that each genre-review pair could be analysed independently. This ensures accurate aggregation and comparison of sentiment metrics across genres.

3. Sentiment Classification:

Sentiment polarity scores were computed using TextBlob and VADER, and reviews were classified based on the following thresholds:



- **Positive:** Score ≥ 0.05
- **Negative:** Score ≤ -0.05
- **Neutral:** Score between -0.05 and 0.05

The resulting dataset is structured to facilitate robust statistical and visual analysis of sentiment distributions, intensity measures, and genre-specific patterns, forming the foundation for the methodology described in the subsequent sections.

4. Methodology

4.1 Sentiment Analysis

In this study, we employed a **hybrid sentiment analysis framework** to systematically quantify audience reactions across movie genres. Two complementary computational approaches were utilized:

1. TextBlob Polarity Scoring:

TextBlob applies a lexicon-based approach to estimate sentiment polarity, producing scores in the continuous range $[-1, +1]$.

Positive values indicate favorable sentiment, negative values indicate unfavorable sentiment, and scores near zero denote neutral sentiment.

This method allows for fine-grained sentiment quantification at the individual review level.

	clean	sentiment
0	laughs overall motivation characters incompreh...	-0.293750
1	waiting exhale waiting waiting waiting waiting...	0.244322
2	angela basset good expected whitney range actr...	0.300000
3	movie okay mediocre might even word describe s...	-0.003889
4	got opportunity see waiting exhale second time...	0.231944

2. VADER Sentiment Scoring:

VADER is a rule-based model optimized for social-media and informal text, capable of detecting intensity-modulating constructs such as negations, degree modifiers, punctuation emphasis, and emoticons.

It outputs compound scores, as well as discrete positive, neutral, and negative proportions for each text instance, enabling robust classification even with short or informal reviews.

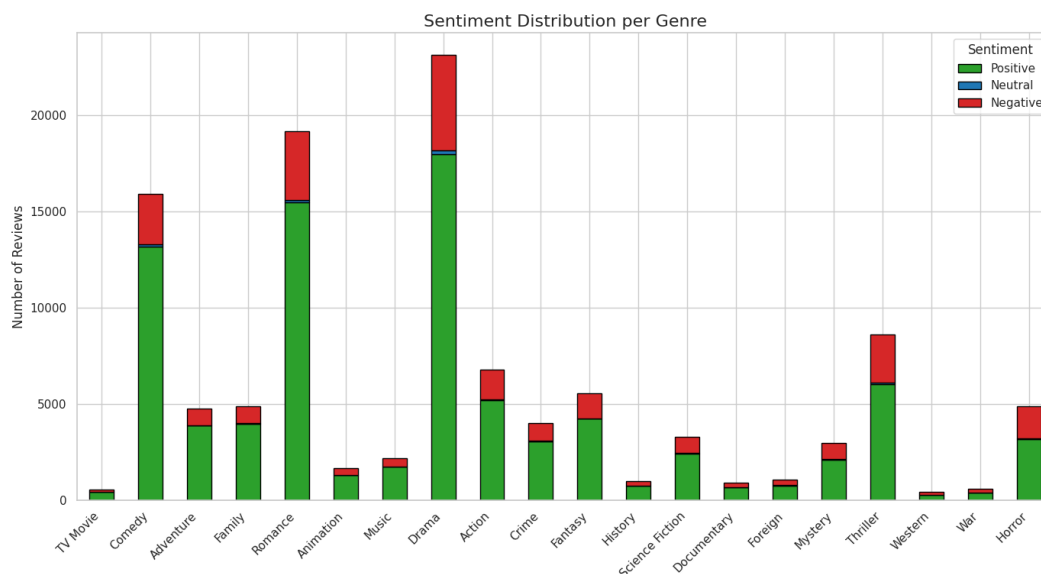
By combining these tools, we ensured both accuracy and robustness of sentiment classification.

4.2 Data Visualization and Exploratory Analysis

We conducted **exploratory data analysis (EDA)** to visualize the distribution and intensity of sentiments across movie genres, using both **aggregate metrics** and **genre-specific metrics**.

1. Sentiment Distribution per Genre

- **Methodology:** Constructed stacked bar plots to depict counts of positive, neutral, and negative reviews per genre.

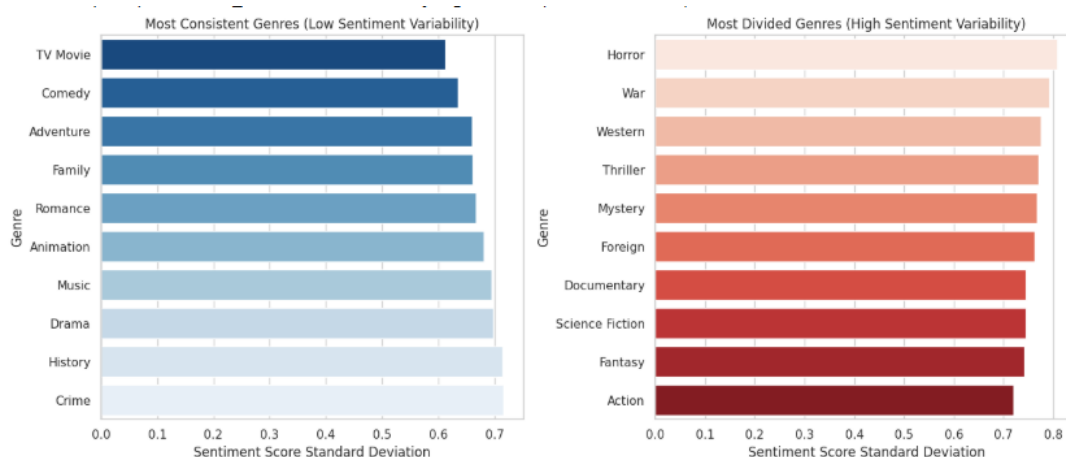


- Observations:**

- TV Movie exhibits the highest proportion of positive reviews (~84%), despite a lower total review count.
- Comedy, Romance, and Drama show substantial review volumes with predominantly positive sentiment.
- Horror and War genres demonstrate elevated negative sentiment ratios, indicating a more critical audience response.

2. Mean Sentiment Intensity per Genre

- Methodology:** Calculated mean polarity/compound sentiment scores to quantify **sentiment magnitude** beyond mere classification.



- Findings:**

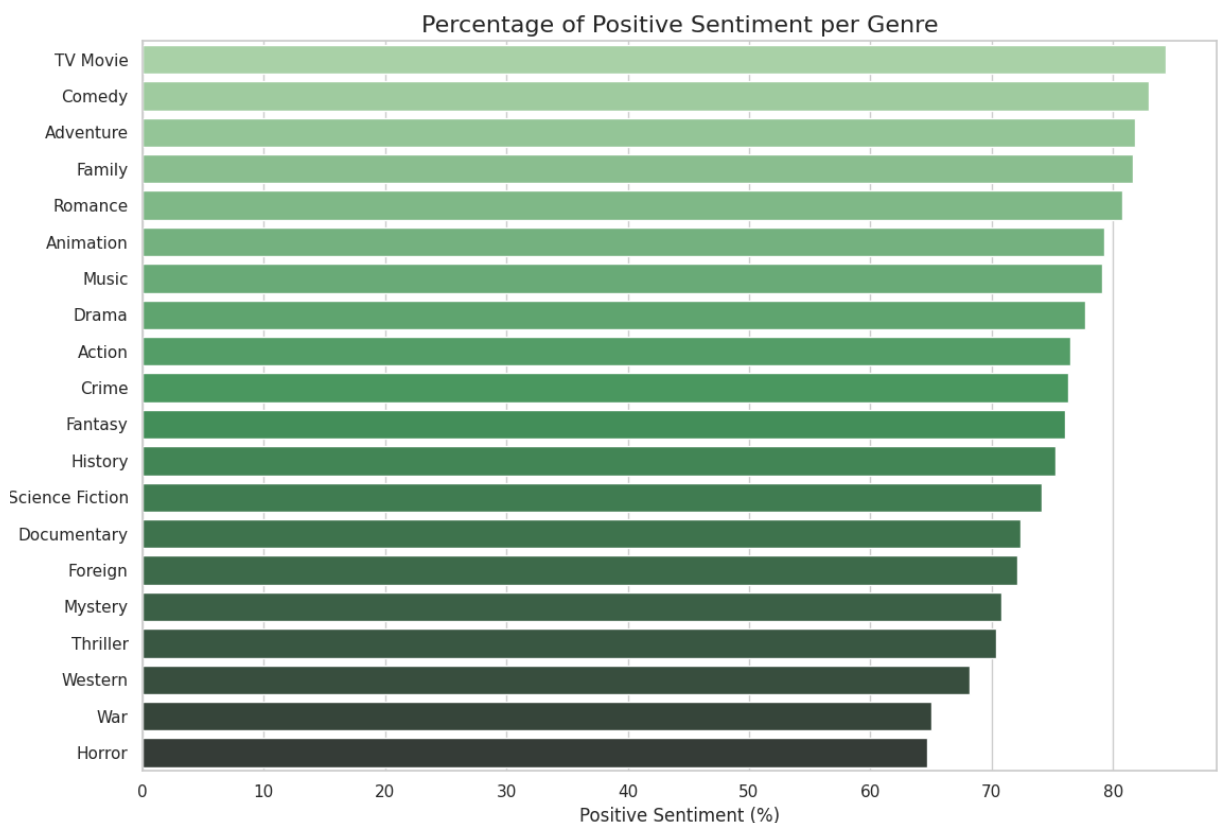
- TV Movie, Comedy, and Adventure genres consistently show high mean scores, reflecting strong audience approval.

- Horror, War, and Western genres show lower mean scores, highlighting polarized or critical sentiment.

By combining **categorical sentiment distribution** and **continuous sentiment intensity**, we gain a comprehensive understanding of both **audience consensus** and **emotional strength** across genres.

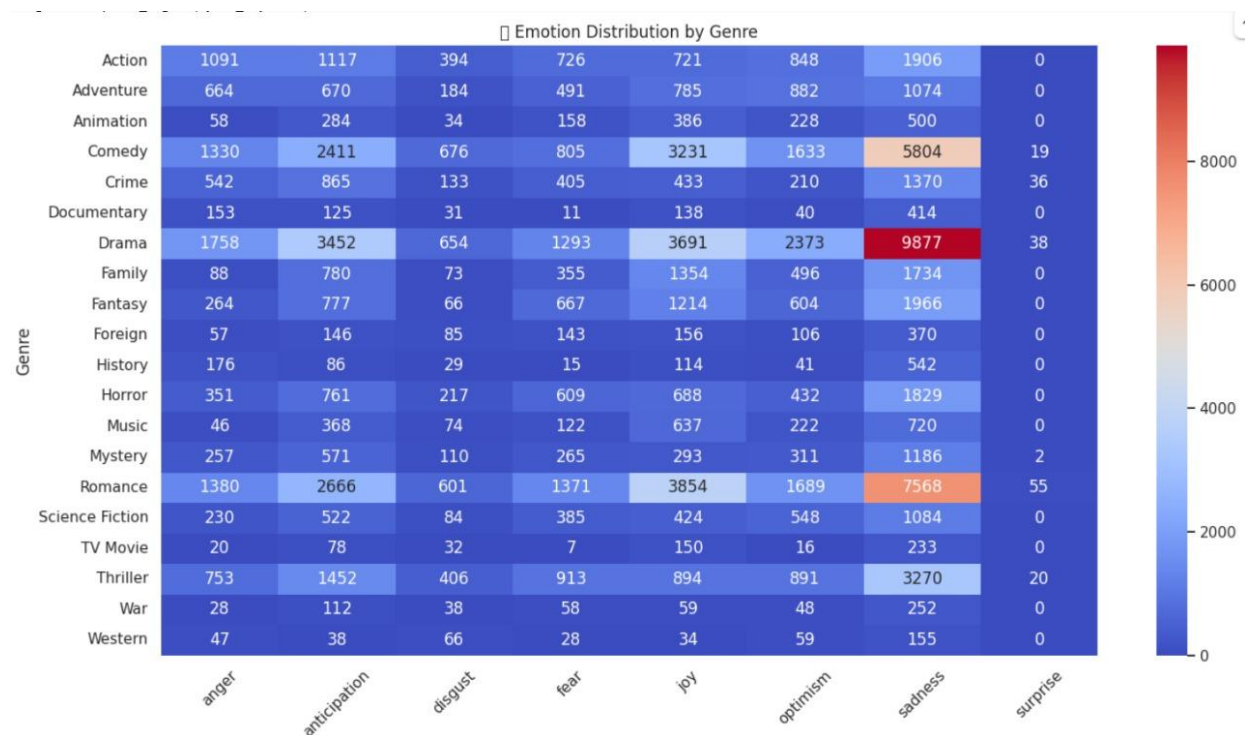
5. Results

5.1 RQ1 – Positive Sentiment Patterns by Genre



- Analysis revealed that **light-hearted genres** (Comedy, Adventure, TV Movie) evoke the highest positive sentiment responses.
- Conversely, **intense or serious genres** (Horror, Thriller, War) elicit higher negative sentiment proportions.
- Combining **percentage of positive reviews** with **mean sentiment intensity** allows us to distinguish between **general approval** and **enthusiastic engagement**, capturing subtleties in audience reaction.

5.2 RQ2 – Sentiment Consistency versus Variability:



1. In this analysis, we visualized the distribution of emotions across different movie genres. Since many movies belong to multiple genres, the data was first exploded so that each movie genre pair is treated as a separate row.
2. Then, we grouped the data by genre and emotion to count how often each emotion occurs within each genre. This grouped data was transformed into a pivot table, where rows represent genres and columns represent emotions. Finally, we plotted a heatmap, where the numbers inside the cells show the exact counts and the color intensity indicates their magnitude.
3. From the heatmap, we can observe trends such as Drama and Romance containing very high sadness counts, Comedy being strongly associated with joy, while other genres like Action and Adventure show a more balanced mix of emotions. This visualization makes it easy to compare emotional patterns across genres at a glance.

5.3 RQ3 – Practical Implications of Sentiment Analysis

- Automated sentiment quantification serves as a **data-driven proxy for audience engagement** and preference mapping.
- Creators and marketers can leverage sentiment insights to **optimize content strategies**, align production with audience expectations, and anticipate reception dynamics.
- This application demonstrates how **computational text mining bridges qualitative human feedback with quantitative analysis**, enhancing decision-making in the film industry.

6. Live Experiments & Real-Time Audience Engagement

A key strength of our poster presentation was the relatability of the research topic. Since movies are a part of everyday life, many audience members were naturally drawn to the discussion and began sharing their favourite genres and personal viewing preferences. This spontaneous interaction created an engaging atmosphere where participants could directly connect their experiences with the sentiment analysis results we presented.

Observations: The discussion revealed how strongly personal taste and emotional association with genres (e.g., Comedy as stress-relief, Horror as thrill-seeking, Drama as thought-provoking) influence the way audiences interpret reviews. This highlighted the subjective dimension of sentiment, which computational models attempt to quantify.

Audience Response: The audience appreciated that the project bridged academic research with real-life experiences, making abstract computational methods more tangible. By linking sentiment scores back to their own favourite genres, participants better understood why sentiment analysis matters — not just as a technical tool, but as a way of capturing collective cultural and emotional trends in entertainment consumption.

7. Reflection

7.1 Poster Development

During the poster preparation and peer-review stage, several critical aspects emerged:

Feedback emphasized the need for cognitive economy in visualization design, stressing that legends, color scales, and labels must remain concise and interpretable without oversimplifying the data.

To enhance analytical depth, we integrated both proportional sentiment distribution metrics (positive/negative/neutral ratios) and central tendency measures of sentiment intensity (mean polarity/compound scores). This multimodal representation ensured that visualizations captured not only the direction of sentiment but also its magnitude and variability, offering a dual-layered interpretive framework.

7.2 Presentation Insights

The interactive component of the presentation provided additional meta-level insights:

Live experiments served as a pedagogical reinforcement mechanism, enhancing audience comprehension by grounding abstract NLP concepts in tangible, real-time outcomes.

Engagement increased when the audience recognized the scalability of sentiment analysis pipelines to domains beyond cinema, such as product reviews, political discourse, and social media monitoring.

Suggestions for future work :

Multilingual Extension: Incorporating cross-lingual and multilingual sentiment models (e.g., multilingual BERT or XLM-R) to broaden the dataset applicability across global markets.

Linguistic Complexity Modeling: Factoring in review length, syntactic richness, and semantic density as potential moderators of sentiment classification accuracy. This aligns with recent work emphasizing the role of stylistic variance and discourse structures in shaping sentiment interpretability.

8. Conclusion

The empirical findings of this study demonstrate that sentiment analysis can serve as a scalable, data-driven proxy for audience reception in the film domain. Our results suggest:

Genre-specific sentiment profiles: Light-hearted genres (Comedy, Adventure, TV Movie) are associated with consistently high positive sentiment scores, while darker or more emotionally demanding genres (Horror, War, Drama) exhibit elevated variability and polarized responses.

Practical Relevance: Sentiment analysis offers actionable insights for filmmakers, distributors, and marketers by identifying preference clusters and risk-prone genres, thereby informing creative and commercial decision-making.

Methodological Contribution: The integration of distributional sentiment measures with intensity-based scoring enables a richer interpretive model than reliance on a single metric. This combined framework distinguishes between weak approval vs. strong endorsement and between mild criticism vs. strong rejection, enhancing interpretive precision.

Ultimately, this study highlights the broader potential of computational text mining in bridging the divide between subjective audience experiences and objective, quantifiable metrics, thereby contributing to both academic research and applied industry practices.

9. Acknowledgements

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10. References

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